Exp 10

Ghidra 11.1 - CodeBrowser: libsample.so [Listing + Decompiler: check_license]

Listing: libsample.so — .text: check_license

```
00101140    55              push    rbp
00101141    48 89 e5        mov     rbp, rsp
00101144    48 83 ec 20     sub     rsp, 0x20
00101148    48 89 7d e8     mov     qword ptr [rbp - 0x18], rdi
0010114c    48 8b 45 e8     mov     rax, qword ptr [rbp - 0x18]
00101150    48 8d 35 b9     lea     rsi, [s_SECRET_LICENSE_KEY]
00101157    48 89 c7        mov     rdi, rax
0010115a    e8 d1 fe ff     call    strcmp
0010115f    85 c0           test    eax, eax
00101161    75 07           jnz     LAB_0010116a
00101163    b8 01 00 00     mov     eax, 0x1 ; return 1 (Success)
00101168    eb 05           jmp     LAB_0010116f
0010116a    b8 00 00 00     mov     eax, 0x0 ; return 0 (Failure)
0010116f    c9              leave
00101170    c3              ret
```

Decompiler: check_license — C Pseudocode

```
/* Decompiled with Ghidra 11.1 */
ulong check_license(char *user_input_key)
{
    int cmp_result;

    // Hardcoded static credential comparison
    cmp_result = strcmp(user_input_key, "SECRET_LICENSE_KEY_2026");
    if (cmp_result == 0) {
        /* License validation successful */
        return 1;
    }
    return 0;
}
```

[*] VULNERABILITY ANALYSIS:

- Hardcoded secret key located in .rodata string pool.
- Bypassed via binary patch (NOP/JMP) or Frida JNI hook.

Administrator: Command Prompt - ROPgadget

```
Microsoft Windows [Version 10.0.22631.4317]
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C:\Users\Manoj\Mobile-Pentesting-Lab> ROPgadget --binary libsample.so --only "ret"
Gadgets information
============================================================
0x0000000000101016 : ret
0x0000000000101034 : pop rbp ; ret
0x000000000010104e : pop rbx ; pop rbp ; ret
0x0000000000101062 : pop r12 ; pop r13 ; pop r14 ; pop r15 ; ret
0x000000000010107a : pop rdi ; ret
0x0000000000101088 : pop rsi ; pop r15 ; ret
0x00000000001010a2 : pop rdx ; ret
0x0000000000101115 : add rsp, 0x18 ; pop rbx ; pop rbp ; ret
0x0000000000101170 : leave ; ret
============================================================
Unique gadgets found: 42

C:\Users\Manoj\Mobile-Pentesting-Lab> echo [*] ROP gadget inventory complete for libsample.so.
```